COMP202 Week 6 - some proofs

This document shows proofs of some properties of graphs. The goal is to familiarise yourself with basic graph-theoretic notation and with writing proofs formally in this language.

Lemma 1. *If G is an undirected graph with m edges, then*

$$\sum_{v \in V} \deg(v) = 2m.$$

Proof. Let (a, b) be an edge. This edge is counted twice in the above summation: once when the vertex being summed over is a, and once for b. Thus, every edge is counted exactly twice, and hence the total sum is twice the number of edges, which is $2m$. $\square$

Lemma 2. *If G is a directed graph with m edges, then*

$$\sum_{v \in V} \text{in-deg}(v) = \sum_{v \in V} \text{out-deg}(v) = m.$$

Proof. Exactly the same proof as the previous one works, with the additional observation that every edge is now counted once instead of twice. $\square$

For all remaining statements, we assume that G is an undirected graph with n nodes and m edges.

Lemma 3. *If G is a tree (doesn't contain a cycle) and it is connected, then $m = n - 1$.*

Proof. We prove this by induction on n.

Base case: The case when $n = 1$ is trivial.

Inductive step: Since G is a tree, it has no cycle, and hence it has a vertex of degree 1 (why?). Let v be such a vertex, and let (u, v) be the edge that is incident on v. Remove the edge (u, v) and the vertex v to obtain a graph G' on $n - 1$ vertices that is also a tree (why?). By the inductive hypothesis, G' has $n - 2$, and so G has $n - 1$ edges. $\square$

Lemma 4. *If G has $m \geq n$ edges, then G contains a cycle.*

Proof. We prove this by induction on n.

Base case: The cases when $n = 1, 2$ trivially hold.

Inductive step: We consider 2 cases:

- If G has a vertex v with $\deg(v) = 1$, then say (u, v) is the edge incident on it. Remove the edge (u, v) and the vertex v to obtain a graph G' with $n-1$ vertices and at least $n-1$ edges. By the inductive hypothesis, G' and hence G contains a cycle.

- In the other case, every vertex has degree at least 2. Here is an informal proof in this case. Start at an arbitrary vertex v_1. "Walk" to a neighbour v_2 of it. Then walk to a neighbour v_3 of v_2. At every step when we see a vertex for the first time, $\deg(v_i) \geq 2$ and so we can walk out of it again. Thus, at some point we must encounter a vertex v_{i+1} that is equal to v_j for some $j < i$, giving us a cycle $(v_j, v_{j+1}, \ldots v_i)$.

$\square$

Lemma 5. *If G has no cycle, then $m \leq n - 1$.*

Proof. This is just the contrapositive form of the previous lemma. $\square$

Lemma 6. *If G is connected, then $m \geq n - 1$.*

Proof. We prove this by induction on m.

Base case: The case when $m = 1$ is easy.

Inductive step: If G is a tree, then an earlier lemma proves that $m = n - 1$, and we are done. If G is not a tree then G has a cycle. Let $G' \subseteq G$ be obtained from G by removing any edge from that cycle. Let n', m' denote the number of vertices and edges in G', respectively. Note that G' is still connected and $m' = m - 1$. By the inductive hypothesis, $m' \geq n' - 1$. But $n' = n$ since we removed no vertex. Thus,

$$m - 1 = m' \geq n' - 1 = n - 1.$$

Thus, $m \geq n$ and this shows the inductive step. $\square$